A PALAEOMAGNETIC AND PETROMAGNETIC STUDY OF UPPER-CARBONIFEROUS, PERMIAN AND TRIASSIC SEDIMENTS, NW BULGARIA

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Summary: *In order to obtain basic palaeomagnetic data on Upper Carboniferous, Permian and Triassic sediments collected from the NW Bulgaria, laboratory stability tests were extended from A.C. and thermal treatments to studies of mineral phase changes and to investigations of changes of magnetic anisotropy during laboratory procedures. Laboratory criteria were found which permitted to distinguish samples suitable for palaeomagnetic analyses from those representing rocks totally or almost totally chemically reworked during their history. Palaeomagnetic directions and pole positions derived from Stephanian, Lower Permian and Triassic rocks from the southern margin of the Moesian Platform are compatible with the values obtained for the tectonically stable North-European Platform.*

1. INTRODUCTION

The purpose of a joint Bulgarian—Czechoslovak geophysical expedition was to carry out palaeomagnetic and petromagnetic investigations of Triassic, Permian and Upper Carboniferous rocks on Bulgarian territory with a view to obtaining basic palaeomagnetic data required for palaeotectonic reconstructions in the marginal parts of the Moesian Platform. The expedition was organized by the Geophysical Institute of the Bulgarian Academy of Sciences, Sofia. The staff of the Palaeomagnetic Laboratory of the Institute of Applied Geophysics in Prague, and of the Geophysical Institute of the Czechoslovak Academy of Sciences in Prague, participated in field and laboratory work. In most cases samples were collected from the Fore-Balkan; only the Permian samples coming from the Kurilo locality were taken from the Balkan. The Triassic, Permian and Upper Carboniferous are all represented by sedimentary rocks, particularly red sandstones and red claystones, rarely red quartzites.

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Extensive laboratory measurements were carried out with the aim to differentiate rocks suitable for palaeomagnetic analyses (after application of magnetic cleaning procedures) from rocks totally or almost totally remagnetized during the geological history. Classical laboratory palaeomagnetic stability tests by means of A.C. and thermal fields were supplemented by studies of mineral phase changes and of magnetic anisotropy during laboratory treatments. It was expected that extending laboratory investigations, especially studies of mineral phase changes and studies of changes of magnetic anisotropy during laboratory thermal treatments, would result in obtaining additional data applicable to judging palaeomagnetic stability of rocks investigated.

2. OUTLINE OF GEOLOGY, COLLECTING OF ORIENTED SAMPLES

Oriented samples of Upper Carboniferous (Stephanian), Permian and Triassic sediments were collected from NW Bulgaria for palaeomagnetic analyses. Studies were concentrated especially on "red beds" dominated by sandstones, claystones and quartzites. The investigated rocks showed an essential difference in petrographical, mineralogical and chemical compositions, as well as in their tecture and structure, all resulting from different palaeotectonic, palaeogeographic, facies and palaeoclimatic development history [1, 2].

Upper Palaeozoic rocks were deposited under various palaeogeographic conditions. An ancient mountain system of the Variscides trending NW—SE was formed in the western part of Bulgaria. Sediments of Stephanian age were laid down in intra-orogenic depressions; diluvial, stream, lacustrine and mud sediments were thus formed. Most depressions developed in grabens bounded by tectonic lines which were activated during the early Permian. Lava locally ascended along these tectonic lines towards the surface. This period was succeeded by an intense peneplanation, at which time clastic rocks were being deposited while the climate changed from humid to arid. Thick sediments layers were accumulated in the Lower Permian (Autunian, Saxonian). In Upper Permian (Thuringian), after other tectonic movements, rocks were deposited especially in deltas.

Triassic sediments developed under quite different conditions [1]. A vast depression filled with water of marine nature was formed over the entire territory of western Bulgaria. Since the Variscan mountain system was already a peneplane, equal sedimentation conditions prevailed not only in the study area but also far to the NE and SW. Isolated islands (e.g. in the surroudings of Belogradchik) or prominent depressions (e.g. near Tserovo) can be interpreted on the peneplaned relief.

The deepest Upper Palaeozoic horizons containing rocks of the „red beds" type occur near the Zverino locality. The following sequences can be seen here (from down to upwards) [3]:

A. *Stephanian:* Zlotishka Series — diluvial conglomerate (80—90 m). Ochindolska Series — coal-bearing clay-terrigenous deposits with alluvial and lacustrine sediments (300 m). Zverino Series — terrigenous beds containing stream, terrace and lacustrine sediments (460 m); samples of 2347 A to 2364 A representing sandstones and aleurolites were collected from the last-named beds.

B. *Lower Permian* (*Autunian*): Bukska Series — terrigenous clayee beds with lacustrine and stream sediments (120 m). Vranska Series — alluvial conglomerate (160 m thick).

The sequence of Upper Palaeozoic rocks observable at the Stakevzi locality is as follows [4]:

A. *Stephanian:* Starchovdolska Series — coal-bearing clay-terrigenous beds (340 m).

B. *Stephanian - Permian transitional zone:* Levishka Series — terrigenous-clay beds containing stream, lacustrine and alluvial (subordinate) sediments (235 m); samples of 2195 A to 2202 A were collected from these beds.

C. *Lower Permian* (*Autunian*): Vranska Series — containing three beds consisting of terrigenous (25—580 m), terrigenous-volcanogenic (20—50 m) and breccia-conglomerate (up to 750 m) rocks.

Table 1. Review of samples of Stephanian — Triassic sediments collected for palaeomagnetic investigations in this study.

Sample Nos.	Age	Lithology	Locality	Notes
95A—111A	Upper Permian	Sandstones	Lesidren	Signs of weathering
2103A—2120A	Lower Permian	Sandstones	Kurilo, road and railway cuttings	Signs of weathering
2121A—2137A	Lowermost Triassic	Sandstones mainly	Smolianovci, section Rikovska bara	Signs of weathering
2138A—2155A	Upper Permian (Thuringian)	Sandstones mainly	Smolianovci, section Rikovska bara	Signs of weathering
2156A—2165A	Lower Permian (Upper "Rotliegendes", Saxonian)	Sandstones mainly	Smolianovci, section Rikovska bara	Signs of weathering
2175A—2194A	Lower Triassic	Sandstones	Belogradchik, separate outcrops	Effects of strong weathering
2195A—2202A	Transion zone of Stephanian to Lower Permian	Sandstones	Stakevzi, Stakevska reka	
2203A—2211A	Lowermost Triassic	Sandstones	Falkovets, outcrops near the road	Effects of strong weathering
2212A—2214A	Upper Permian (Thuringian)	Sandstones	Falkovets, outcrops along a river	Signs of weathering
2215A—2271A	Lower Permian	Red claystones mainly	Smolianovci, section along a river, numerous outcrops	Compact samples with haematite pigment
2272A—2284A	Triassic	Claystones with specularite	Tserovo, road cutting	Subhorizontal layers

Sample Nos.	Age	Lithology	Locality	Notes
2285A—2291A	Triassic	Claystones	1 km north of Tserovo, road cutting	Subhorizontal layers
2292A—2300A	Triassic	Claystones and sandstones	2 km north of Tserovo, road cutting	Subhorizontal layers
2301A—2305A	Lower Triassic	Sandstones	Bov, quarry near a road	Subhorizontal layers
2306A—2316A	Lower Triassic	Sandstones	Between Bov and Lakatnik	Subhorizontal layers
2317A—2328A	Lower Triassic	Sandstones	0·5 km north of Lakatnik, road cutting	Subhorizontal layers
2329A—2340A	Lower Triassic	Sandstones	1·9 km north of Lakatnik, road cutting	Subhorizontal layers
2341A—2346A	Lower Triassic	Sandstones	3·8 km north of Lakatnik, road cutting	Subhorizontal layers
2347A—2364A	Stephanian	Quartzites with haematite pigment	Zverino	Compact samples
2365A—2373A	Lower Permian	Sandstones	Kurilo, right bank of river	Effects of strong weathering
2374A—2385A	Lower Permian	Sandstones	Kurilo, approx. 150 m south of previous exposure	Effects of strong weathering

The Lower Permian (Autunian) sediments found north of Kurilo developed on southern slopes of the Variscan orogenic system [5]. The following series can be observed from bottom to top: Lokorska Series — coarse grained beds with stream and alluvial sediments (200 m). Voynyagovska Series — terrigenous-volcanogenic beds with tuffs, xenotuffs and tephritoids (25—100 m). Podgumerska Series — terrigenous beds with predominant alluvial sediments (600—720 m); samples of 2365A to 2373A were taken from its topmost horizons. Kurilo Series —terrigenous-clay beds containing sediments of continental basins (180 m); samples of 2103A to 2120A and of 2374A to 2385A were collected from the series just mentioned.

Many samples are representative of Permian sediments derived from piedmont basins north of the Variscan orogenic belt. Rock sections of the lower "Rotliegendes" (specifically Upper Autunian) are exposed between Vinishche and Falkovets and in the Smolianovci area [6]. These include the Smolianovska Series subdivided into four members: 1. Perozkiy Member — breccia (1—80 m). 2. Giurgichkiy Member — volcano-tephritoid Series (200—250 m). 3. Karachishkiy Member — polymictic sandstone series (200—300 m). 4. Koritarskiy Member — a series of fine grained alluvial and lacustrine sediments (100—600 m) from which samples of 2215A to 2271A were collected. The following rocks can be ranged to the upper „Rotliegendes" (Saxonian): Vranska Series — breccia conglomerate (150—1000 m), from which samples of 2156A to 2165A were taken. Above these beds rocks of the Upper Permian (Thuringian) are discordantly layered: Rikovska Series — sandstone series (10—430 m) where samples of 2138A to 2155A (Rikovska bara) and of 2212A to 2214A (Falkovets area) were taken.

The samples were intentionally collected from entire sections (with Permian rocks grading into Triassic ones) and from typical occurrences of Triassic rocks in order to check in a laboratory way alteration effects on the changes in magnetic and palaeomagnetic characteristics. The former group comprises sandstone samples of 2121A to 2137A (Rikovska bara) and of 2203A to 2211A (Falkovets area). Samples of 2175A to 2194A represent coarse grained clastics of Lower Triassic age from the aforementioned island near the Belogradchik locality. Other smaller islands yielded samples of 2317A to 2328A (north of Lakatnik). Samples of 2329A to 2346A (Lakatnik area) and of 2301A to 2316A (Bov and Lakatnik areas) represent fine grained sandstones, aleurolites and argillites. Finally, samples of 2272A to 2300A collected from the Tserovo locality area represent red sediments deposited under tranquil conditions in a Triassic basin.

The main tectonic movements determined in western Bulgaria by biostratigraphical and lithologic methods can be placed in the following periods: 1. The Asturian phase between the Lower and Upper Autunian. 2. Between the Lower and Upper "Rotliegendes" (i.e. between the Autunian and Saxonian). 3. Between the Lower and Upper Permian, close to the Saxonian/Thuringian boundary (Saalic phase). 4. Between the Upper Permian and Lower Triassic (Pfalzian phase).

The rocks were sampled so a s to be representative as much as possible of Triassic, Permian and Upper Carboniferous outcrops over a wider area. The rocks of many outcrops, especially Triassic porous sandstones at Belogradchik, Upper Permin sandstones from Lesidren, Permian porous sandstones from the Kurilo area, and the sequence of Permo-Triassic rocks occurring along the Rikovska bara stream in the Smolianovci area have all shown sings of strong weathering. It is known from previous experiments [7] that the rocks suitable for palaeomagnetic investigations are especially those represented by pelitic sediments, e.g. claystones with haematite pigment, in which the clay component acts as an insulator against effects of exogenous agents. The rocks possessing the mentioned properties have been found along some extensive sections, for instance, in the subhorizontal Triassic beds in the Tserovo—Bov—Lakatnik area and on numerous outcrops of gently dipping rocks of Lower Permian age in the Smolianovci area. The Stephanian red quartzites from the Zverino locality were also found suitable for palaeomagnetic studies. Table 1 summarizes the samples collected and gives a basic description of the sampling sites and lithology.

3. LABORATORY STABILITY TESTS BY MEANS OF A.C. AND THERMAL FIELDS

Palaeomagnetic stability was studied by laboratory methods using A.C. and thermal demagnetization as described elsewhere [8]. The two methods were employed by compensating the outer magnetic field. A.C. demagnetization was carried out by letting the samples rotate round two mutually perpendicular axes with the 5 : 4 angular velocity ratio; the maximum amplitude of the A.C. field reached a value of $1400 \,.\, 1000/4\pi \,[\tilde{\mathrm{A}} \,.\, \mathrm{m}^{-1}]$. Thermal demagnetization proceeded until the Curie point for a particular mineral fraction has been attained. The pertinent graphs show M_0 denoting the magnitude of the modulus of natural remanent magnetic moment, $M_{\tilde{H}}$ – the modulus of the moment after application of A.C. field $\tilde{H}$, and M_t – the value of the moment after thermal demagnetization at temperature t. All the samples were measured on JR-3 and JR-4 high-sensitivity spinner-magnetometers [9].

Representative samples of each outcrop were studied, the specimens collected from the outcrops displaying visible exogenous alteration effects and specimens from the outcrops in deeper river and stream cuts were examined separately. It has been found that the samples of especially porous sandstones differ in their properties from those of red sediments with a higher amount of pelitic fraction.

The samples pf Upper Permian sandstones collected from a long section at Lesidren, northern Bulgaria, clearly showed effects of weathering and, as has been proved by

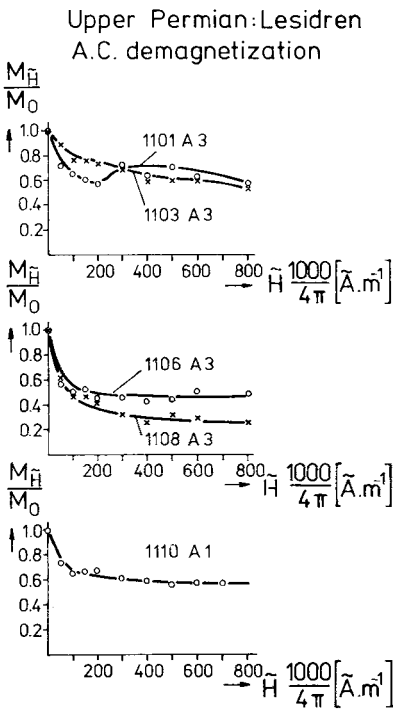


Fig. 1. A.C. demagnetization of Upper Permian red sandstones from Lesidren.

laboratory methods, those coming from lower horizons were weathered as well (Figs 1, 2). The whole population of samples 95A–111A is marked by considerable changes in the directions of remanent magnetic polarization[1]) during thermal de-

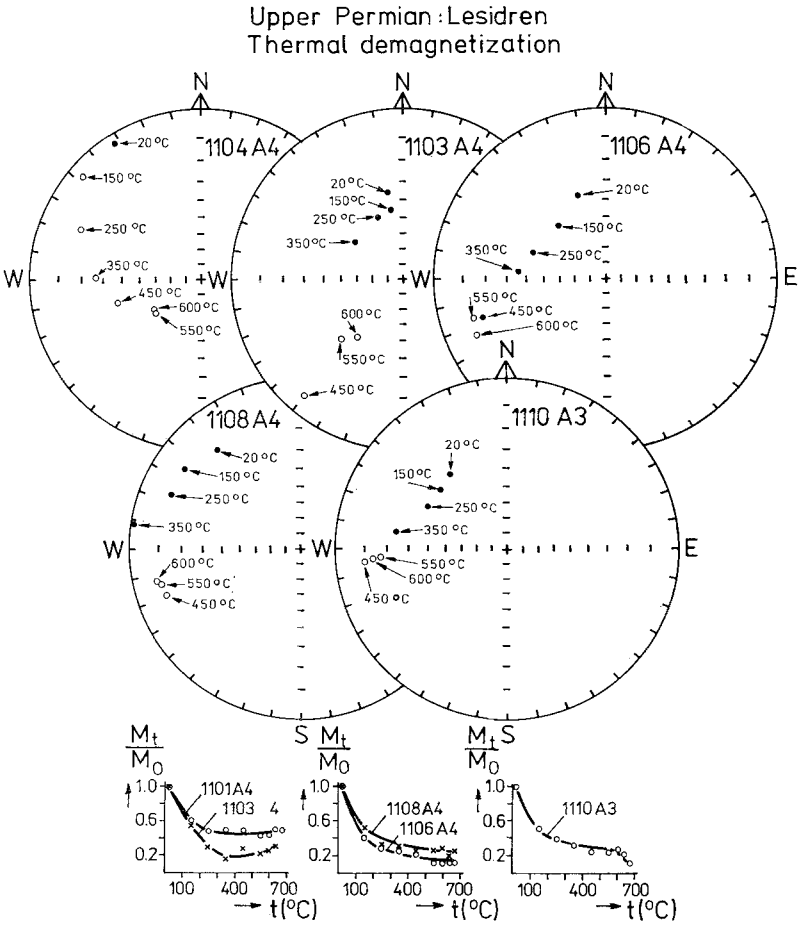


Fig. 2. Thermal demagnetization of Upper Permian red sandstones from Lesidren.

magnetization (Fig. 3, Tab. 2). Although the moduli of RMP decreased considerably close to the Curie temperature for haematite, magnetic cleaning of strongly oxidized samples from Lesidren was not sufficiently effective. Moduli of RMP of the samples in natural state and after magnetic cleaning in the thermal fields are shown in Fig. 4.

The same problem of chemical remagnetization of porous sandstones was encountered in examining Lower Triassic sandstones (2121A–2137A) taken from along the Rikovska bara stream – cf. Fig. 5. Representative samples of porous

[1]) In further text remanent magnetic polarization and natural remanent magnetic polarization are abbreviated as RMP and NRMP, respectively.

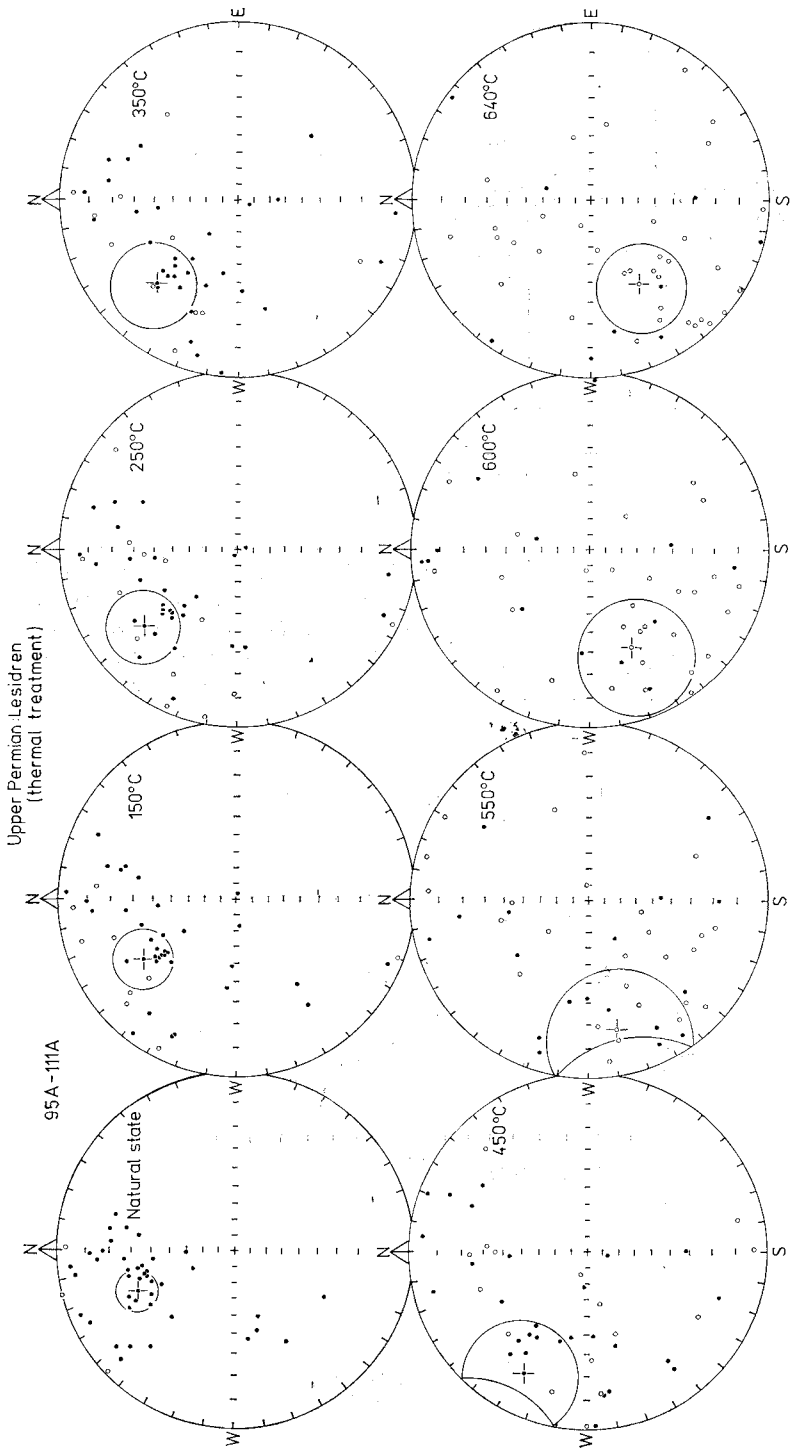


Fig. 3. Directions of NRMP and of RMP of thermally treated Upper Permian red sandstones from Lesidren.

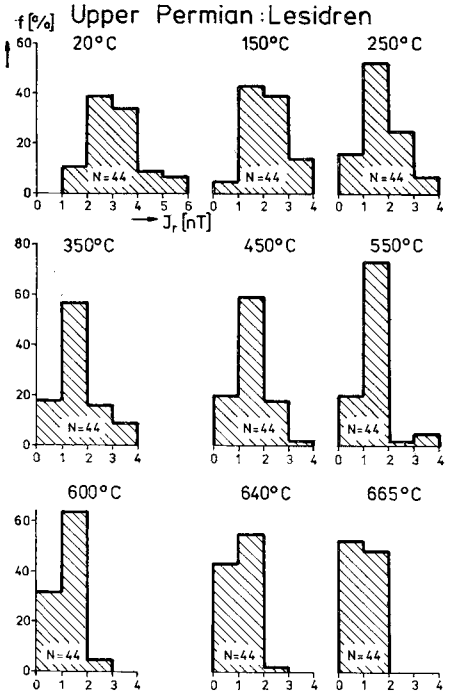


Fig. 4. Histograms of moduli of remanent magnetic polarization of natural and thermally treated Upper Permian red sandstones from Lesidren, 95A—111A. N — number of samples analyzed; f — frequency [inpercentage]; J_r — modulus of RMP in nanotesla [nT].

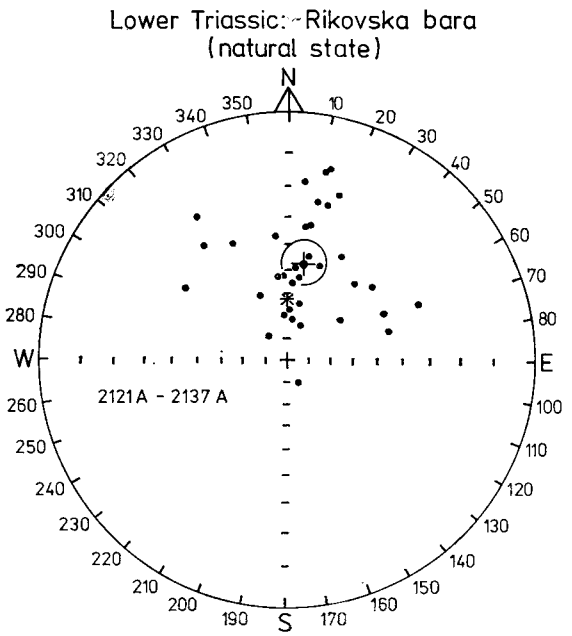


Fig. 5. Directions of NRMP of unstable red sandstones, Lower Triassic, from Rikovska bara.

Triassic and Permian sandstones from other localities also showed remarkable changes in the directions of RMP during thermal demagnetization, see Fig. 6.

Table 2. Thermal treatment of collections of Upper Permian sediments from Bulgaria: Lesidren. Sample Nos.: 95A—111A.

t °C	Mean directions of remanent magnetic polarization		α_{95}	k	N
	D	I			
20	337·2°	29·8°	9·4°	6·3	44[1])
150	326·5°	26·9°	13·2°	3·6	44
250	320·4°	21·9°	16·4°	2·8	44
350	314·2°	22·7°	18·3°	2·4	44
450	296·8°	14·5°	23·4°	1·8	44
550	257·7°	—16·6°	29·2°	1·5	44
600	246·5°	—28·3°	27·4°	1·6	44
	252·2°	—27·8°	28·9°	1·5	44[2])
640	239·5°	—33·0°	21·6°	2·0	44
	244·7°	—37·7°	23·2°	1·9	44[2])

[1]) Natural state.
[2]) Opposite orientation of samples in the furnace.

Different results were obtained from subhorizontally layered Triassic sediments containing interbeds of red sandstones and claystones, in the region of Tserovo, Bov and Lakatnik. A.C. demagnetization disclosed a high magnetic hardness of representative samples under study (Fig. 7); indistinct changes were noted in the directions of RMP during thermal demagnetization. The graphs showing thermal demagnetization indicate that haematite acts as a carrier of palaeomagnetization (Fig. 8). Table 3 showing statistical parameters for thermal demagnetization indicates that declination values increase with increasing temperature. For this reason, the complete population of samples taken from eight outcrops was subjected to magnetic cleaning in the thermal field of 400 °C. Figures 9 and 10 illustrate measuring results

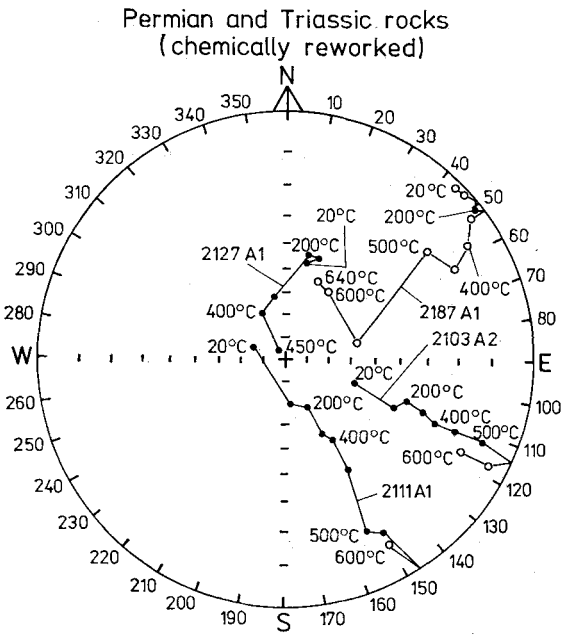


Fig. 6. Directions of NRMP and of RMP of thermally treated palaeomagnetically unstable samples of Permian and Triassic porous sandstones from Kurilo (Lower Permian, 2103A2, 2111A1), Smolianovci, Rikovska bara (Lower Triassic, 2127A1), Belogradchik (Lower Triassic, 2187A1).

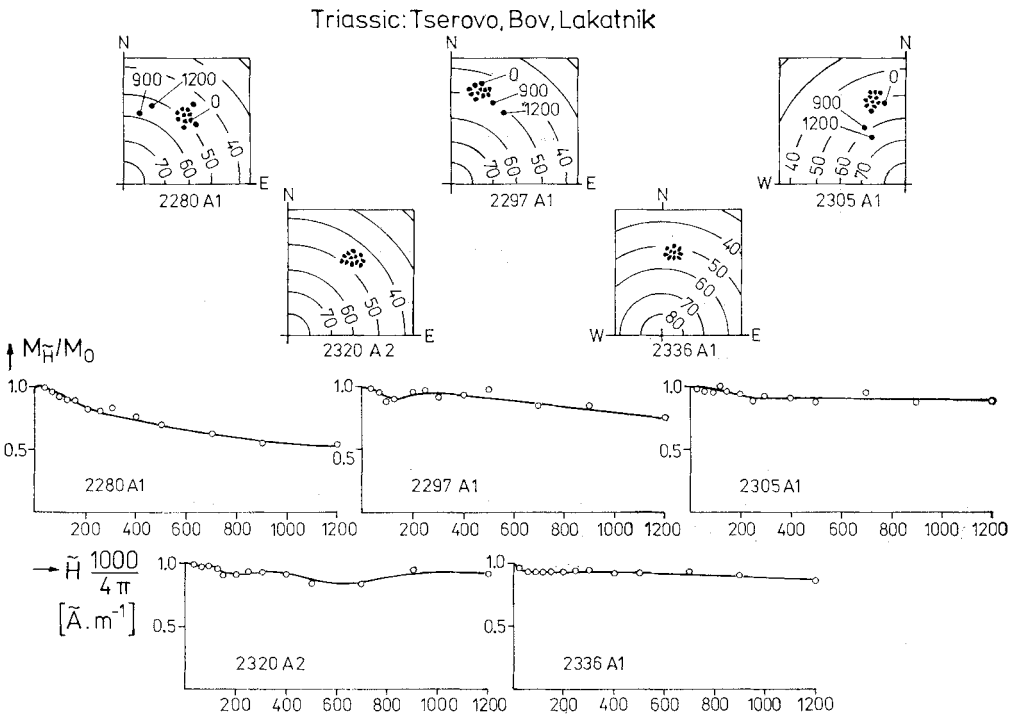


Fig. 7. A.C. demagnetization of Triassic red claystones and sandstones from subhorizontal layers in the region of Tserovo, Bov, Lakatnik, cf. Tab. 1.

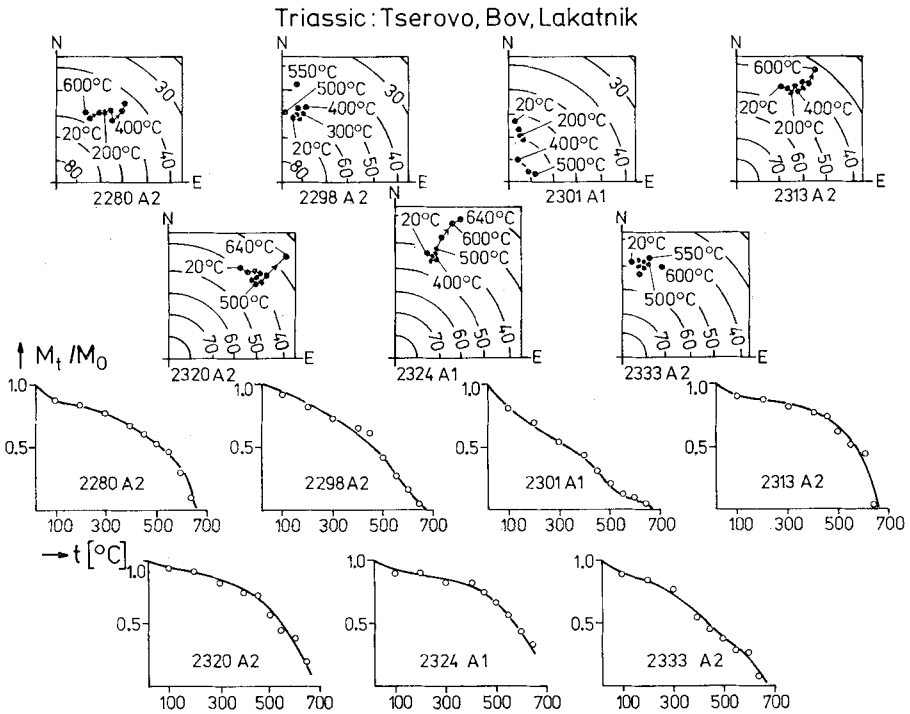


Fig. 8. Thermal demagnetization of Triassic red claystones and sandstones from subhorizontal layers in the region of Tserovo, Bov, Lakatnik, cf. Tab. 1.

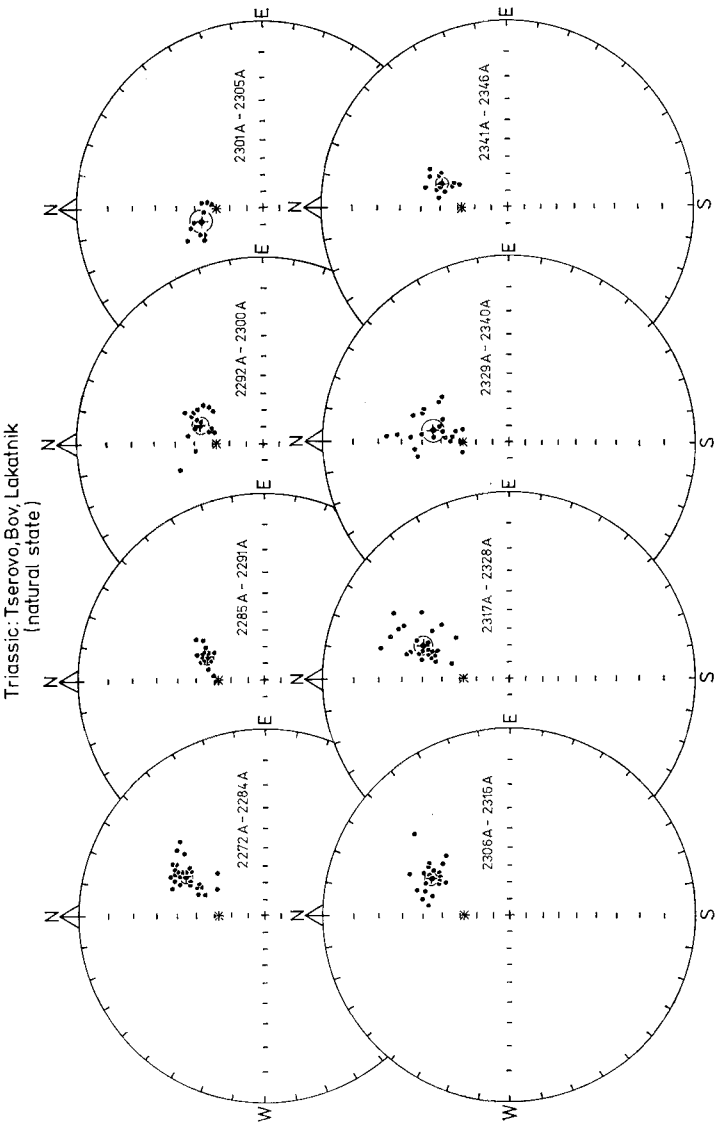


Fig. 9. Directions of NRMP of Triassic sediments from eight sections in the region of Tserovo, Bov and Lakatnik, cf. Tab. 1.

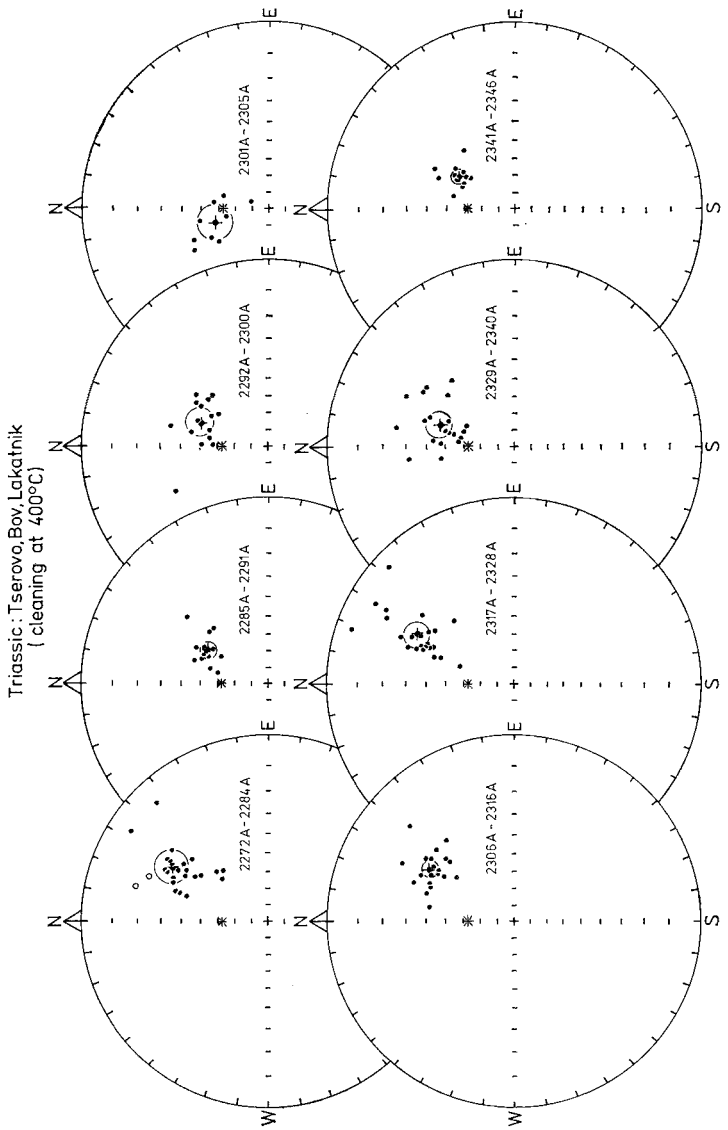


Fig. 10. Directions of RMP of thermally treated (400 °C) Triassic sediments from eight sections in the region of Tserovo, Bov and Lakatnik, cf. Tabs 1 and 4.

Table 3. Thermal tests of representative samples of Triassic sediments from Bulgaria: Tserovo, Bov, Lakatnik. Samples Nos.: 2280A2, 2298A2, 2301A1, 2313A2, 2320A1, 2324A1, 2333A2.

t °C	Mean directions of remanent magnetic polarization		α_{95}	k	N
	D	I			
Natural state	19·26°	50·18°	8·63°	49·85	7
100	23·78°	51·01°	9·12°	44·77	7
200	27·17°	51·00°	.0·05°	37·01	7
300	25·52°	50·25°	10·70°	32·76	7
400	26·40°	52·98°	12·81°	23·16	7
450	30·12°	51·94°	13·41°	21·21	7
500	31·65°	49·55°	14·12°	19·24	7
550	31·74°	49·55°	21·45°	8·87	7
600	38·55°	53·44°	34·78°	3·96	7
640	scattered results				

Table 4. Thermal cleaning of collections of Triassic sediments from Bulgaria: Tserovo, Bov. Sample Nos.: 2272A—2346A.

t °C	Mean directions of remanent magnetic polarization		α_{95}	k	N
	D	I			
Natural state	20·19°	43·09°	1·86°	44·45	133
400	26·07°	39·35°	2·63°	23·33	129
500	27·21°	34·38°	3·36°	14·68	129

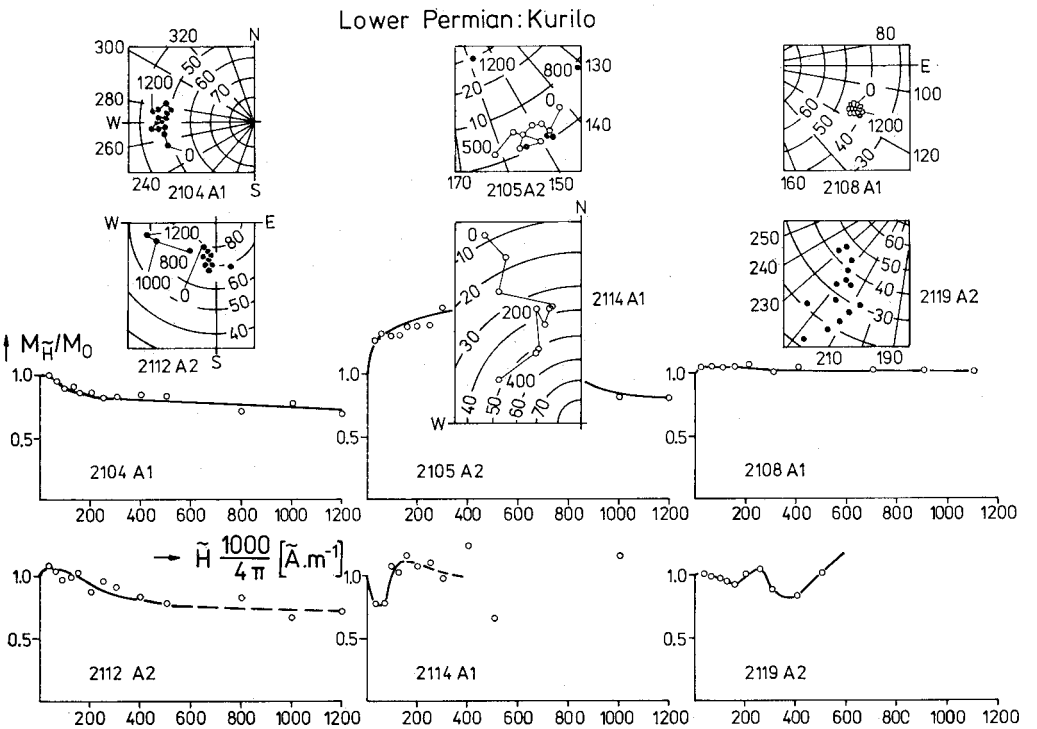


Fig. 11. A.C. demagnetization of representative samples of unstable Lower Permian sandstones from Kurilo.

obtained for the samples in natural state and after thermal cleaning[2]). This complete population was also thermally cleaned in the field of 500 °C. Table 4 indicates that optimum cleaning can probably be achieved in the field of about 400 °C.

Representative samples of Permian sandstones from Kurilo and Rikovska bara are palaeomagnetically unstable, apparently, due to the chemical alteration resulting from weathering – cf. Figs 11, 12.

Quite a different situation appeared in the samples of Lower Permian red claystones collected from five extensive sections stretching along a river cut near the village of Smolianovci. These samples displayed a remarkable magnetic hardness

[2]) A stereographic projection of RMP after cleaning in the A.C. or thermal field and of NRMP is used in this paper. The projection onto the lower (upper) hemisphere is indicated by solid (open) circle. Mean direction calculated using R. Fisher's statistics [10] is indicated by a cross over solid (open) circle and is circumscribed by the circle of confidence at the probability level of $P = 0{\cdot}05$. Direction of the present-day, coaxial, and theoretical magnetic dipole field is denoted by an asterisk. In the Tables (2—7, 9), α_{95} is the semi-vertical angle of the cone of confidence for $P = 0{\cdot}05$, k is the precision parameter, and N indicates the number of specimens analyzed, cf. [11].

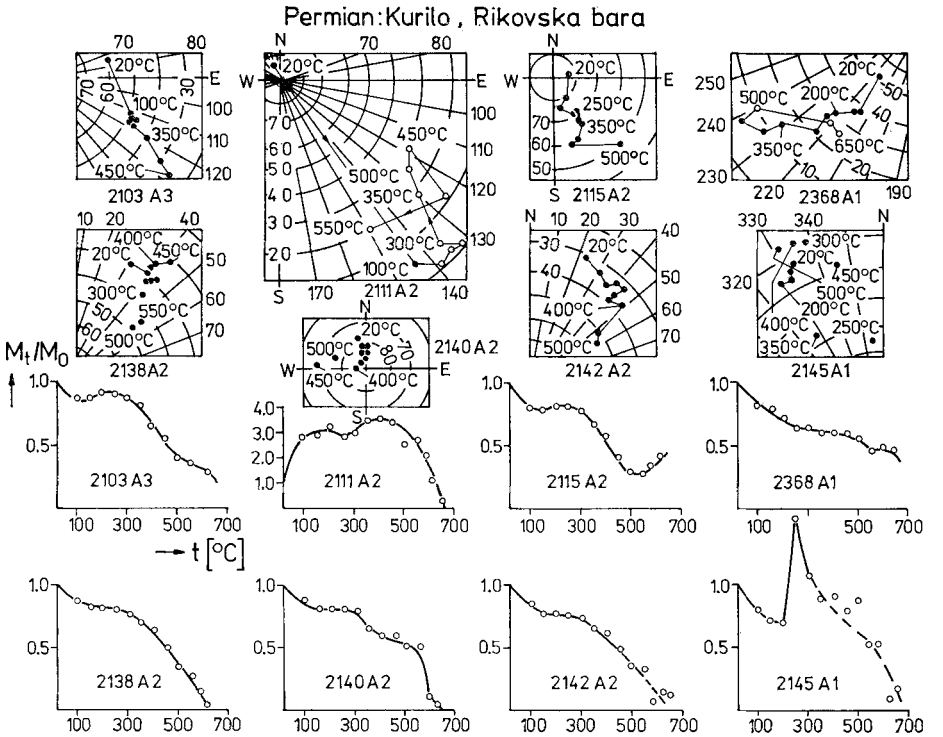


Fig. 12. Thermal demagnetization of representative samples of unstable Permian sandstones from Kurilo and Rikovska bara.

(Fig. 13) and almost generally only small changes in the directions of RMP of thermally demagnetized specimens (Fig. 14). Haematite carries NRMP in most samples. As can be seen from Table 5, the most suitable statistical parameters are those established for the samples thermally demagnetized within the temperature interval of 250–400 °C. Directions of NRMP and RMP after thermal cleaning of the samples at 400 °C are shown in Figs 15 and 16, respectively. The efficiency of thermal cleaning can be disclosed by comparing the two figures and can well be inferred from the statistical parameters given in Tab. 6.

The Stepahian – Lower Permian red sandstones from Stakevzi (2195A–2202A) show a considerable scatter in the directions of NRMP (Fig. 17). The changes in the directions of RMP during A.C. demagnetization required additional demagnetization in the fields higher than $1400 \,.\, 1000/4\pi$ [$\tilde{A} \,.\, m^{-1}$]. Owing to the high magnetic hardness (Fig. 18), it was necessary to use thermal demagnetization. However, since most of the samples underwent phase changes and the thermal cleaning thus failed to be efficient (Fig. 19), the entire population of samples of Stephanian – Lower Permian sandstones from the Stakevzi locality was omitted from further palaeomagnetic considerations. The samples of Stephanian quartzites pigmented with

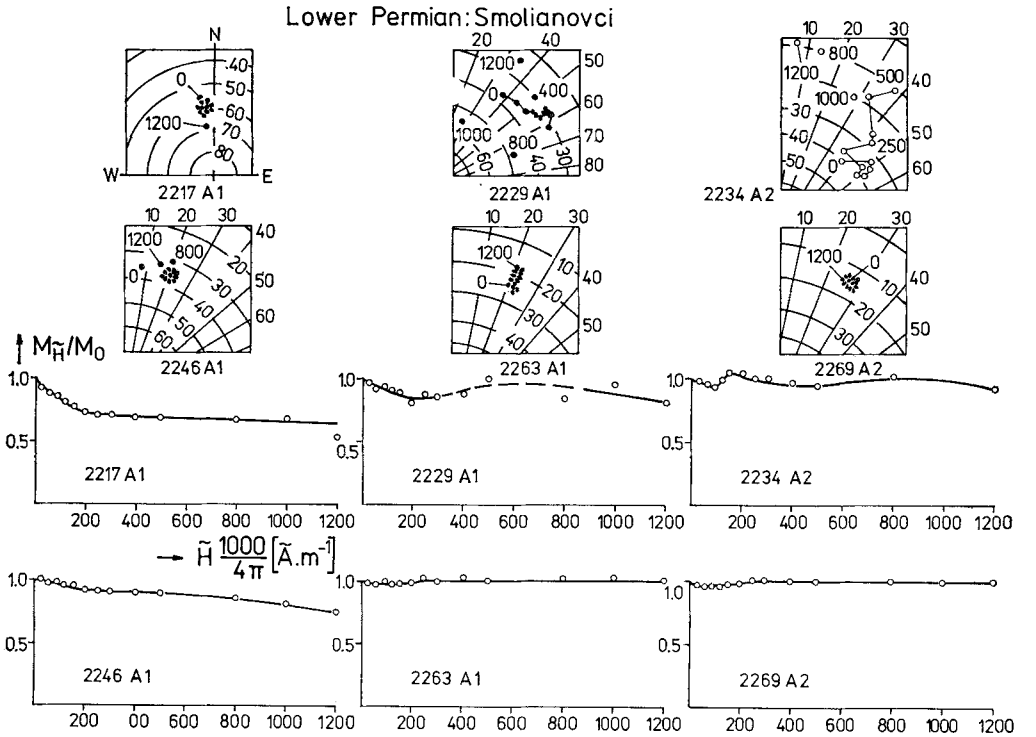


Fig. 13. A.C. demagnetization of representative samples of stable Lower Permian claystones from sections at Smolianovci.

haematite and collected from the locality of Zverino show properties suitable for palaeomagnetic investigations. Representative samples displayed a strikingly high magnetic hardness (Fig. 20) and relatively small changes in the directions of RMP (Fig. 21) were observed during thermal demagnetization until temperatures below the Curie point for haematite were attained. The whole population of samples (2347A – 2364A) was therefore magnetically cleaned in the 400°, 500° and 600 °C fields (Fig. 22). Optimum statistical parameters were obtained at a temperature of 400 °C, so that the directions of RMP of the samples cleaned in the 400 °C field can be regarded as representative palaeomagnetic directions (Tab. 7).

4. STUDY OF MAGNETIC ANISOTROPY DURING LABORATORY STABILITY TESTS

The magnetic susceptibility of the sedimentary rocks investigated in natural state ranges from 20·0 to 230·0 $\times 10^{-6}$ [SI], their geometric mean $\bar{\varkappa}$ being = 91·3 $\times$ $\times 10^{-6}$ [SI] and standard deviation $\varrho_{\ln\varkappa} = 0{\cdot}338$.

Magnetic susceptibility anisotropy attains a low value in most samples. The statistical F-test performed at the confidence level of $p = 0{\cdot}05$ has revealed that

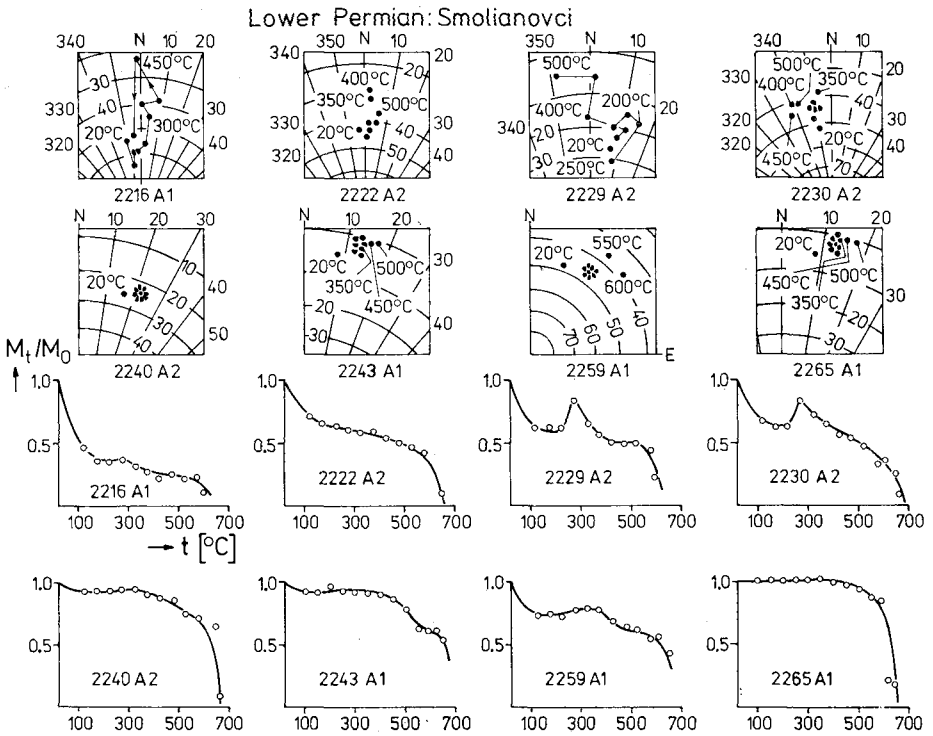


Fig. 14. Thermal demagnetization of representative samples of stable Lower Permian claystones from sections at Smolianovci.

Table 5. Thermal tests of representative samples of Lower Permian sediments from Bulgaria: Smolianovci. Sample Nos.: 2216A1, 2222A2, 2229A2, 2230A2, 2231A1, 2233A1, 2235A1, 2237A2, 2240A2, 2243A1, 2259A1, 2265A1.

t °C	Mean directions of remanent magnetic polarization		α_{95}	k	N
	D	I			
Natural state	12·72°	26·52°	17·05°	7·44	12
100	21·12°	17·11°	28·68°	3·26	12
150	22·48°	13·85°	26·15°	3·72	12
200	23·03°	13·32°	25·01°	3·98	12
250	13·46°	39·30°	11·07°	16·35	12
300	15·45°	31·42°	10·72°	17·35	12
350	17·46°	29·29°	11·62°	14·93	12
400	16·86°	28·99°	11·96°	14·15	12
450	14·75°	25·19°	12·84°	12·40	12
500	18·53°	30·50°	14·96°	9·38	12
550	24·60°	28·72°	17·00°	7·48	12
580	36·36°	15·36°	16·86°	7·59	12
620	82·33°	3·78°	30·82°	2·95	12

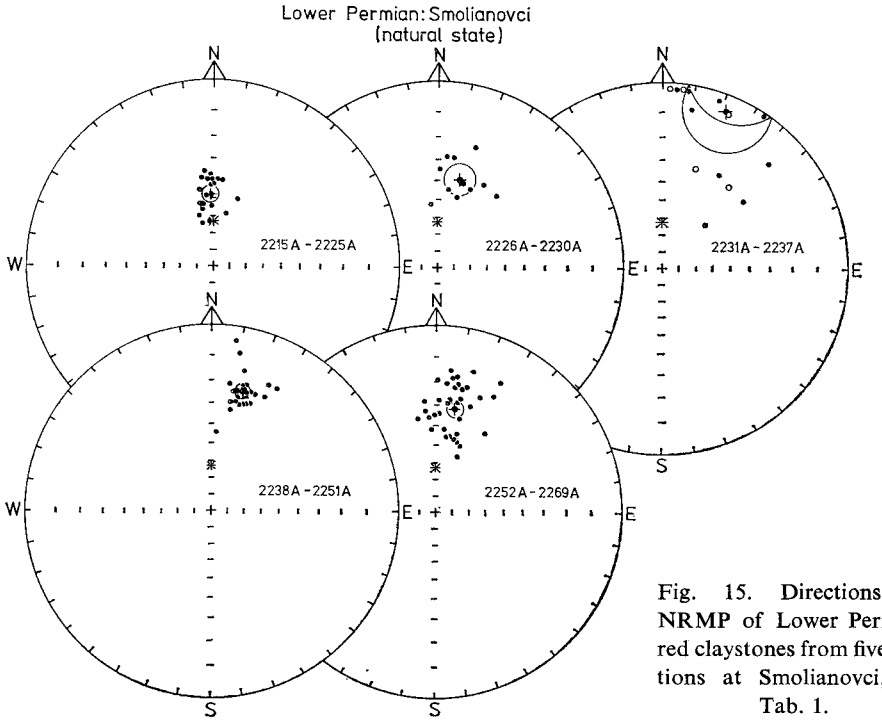


Fig. 15. Directions of NRMP of Lower Permian red claystones from five sections at Smolianovci, cf. Tab. 1.

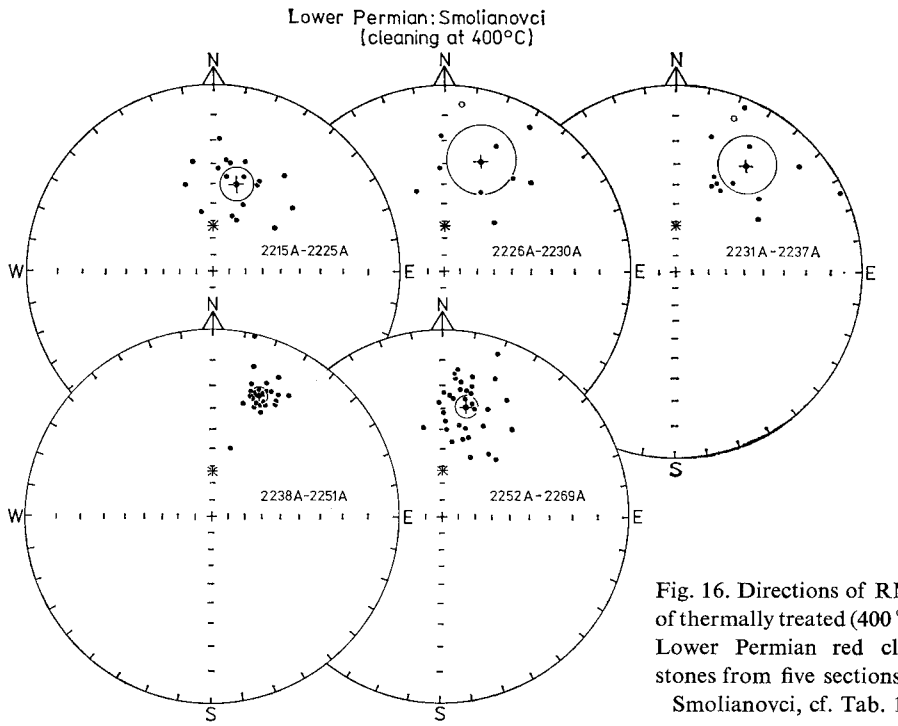


Fig. 16. Directions of RMP of thermally treated (400 °C) Lower Permian red claystones from five sections at Smolianovci, cf. Tab. 1.

Table 6. Thermal cleaning of collections of Lower Permian sediments from Bulgaria: Smolianovci. Sample Nos.: 2215A—2269A.

t °C	Mean directions of remanent magnetic polarization		α_{95}	k	N
	D	I			
Natural state	11·40°	30·12°	3·48°	16·56	106
250	15·09°	28·85°	4·01°	13·44	100
300	17·98°	30·89°	3·91°	14·05	100
350	18·34°	28·75°	3·56°	16·73	100
400	18·72°	27·15°	3·32°	19·13	100

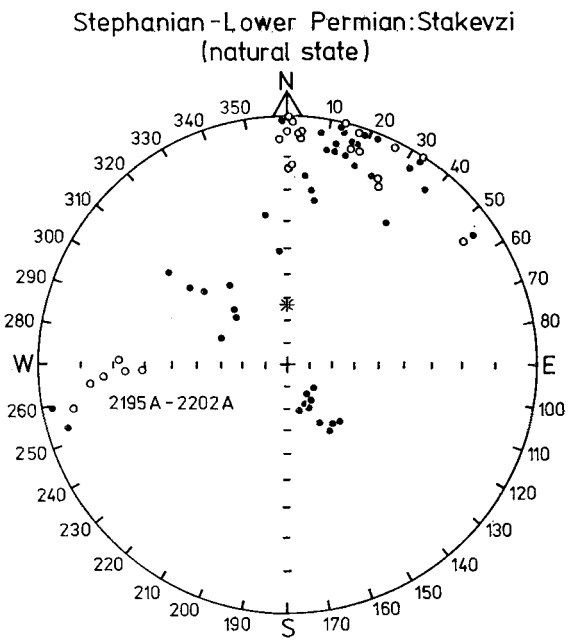


Fig. 17. Directions of NRMP of unstable Stephanian — Lower Permian red sandstones from Stakevzi, cf. Tab. 1.

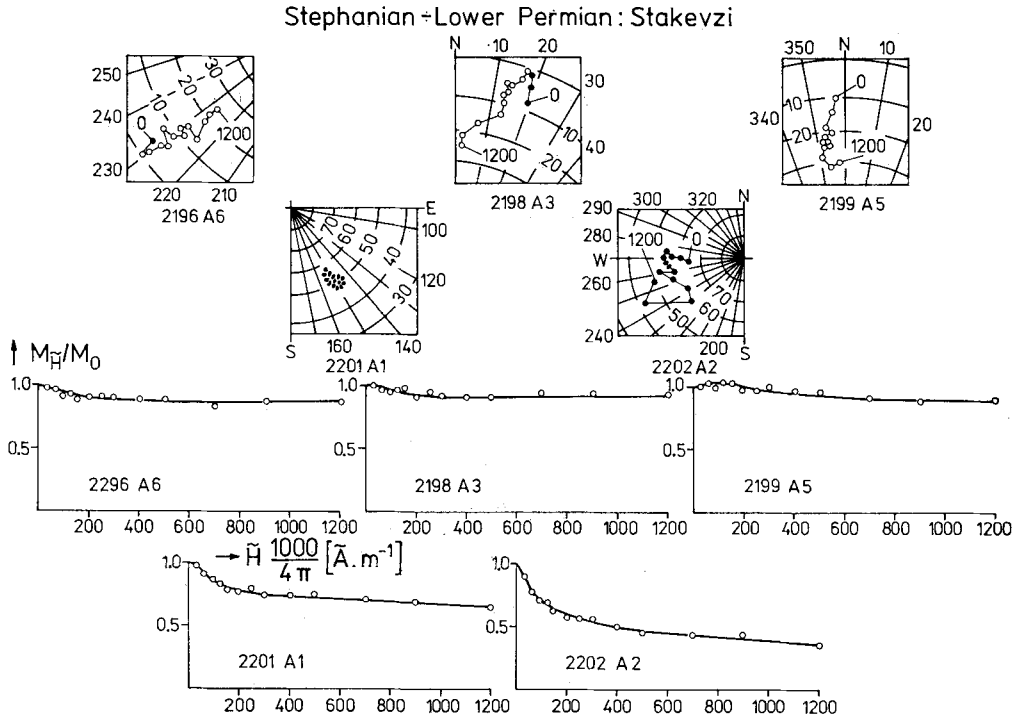


Fig. 18. A.C. demagnetization of Stephanian — Lower Permian red sandstones from Stakevzi.

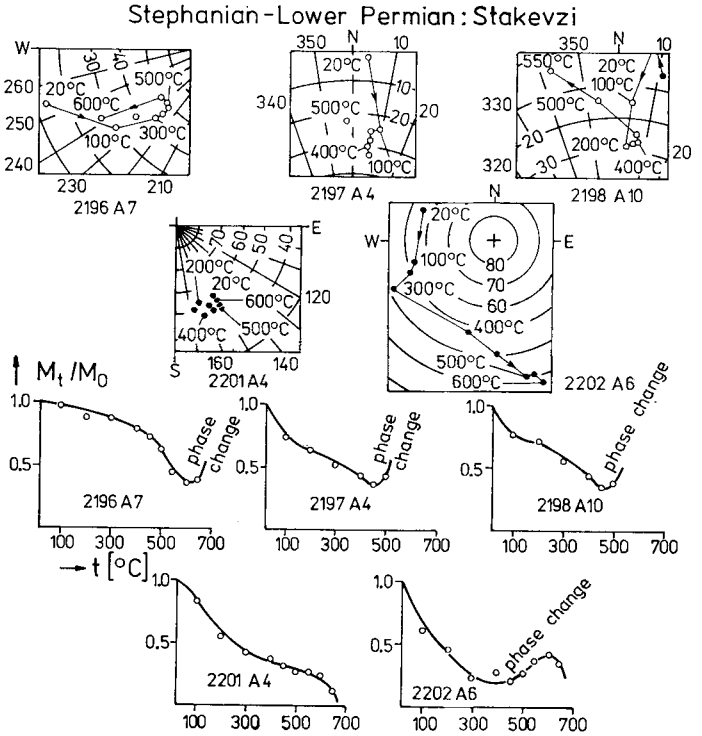


Fig. 19. Thermal demagnetization of Stephanian — Lower Permian red sandstones from Stakevzi.

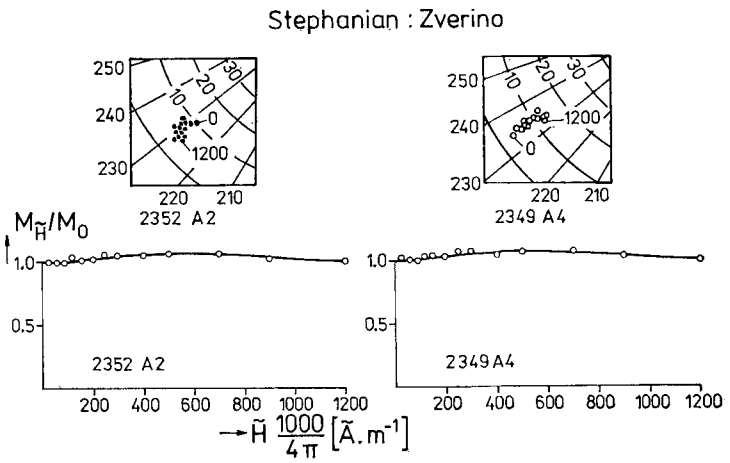


Fig. 20. A.C. demagnetization of Stephanian red quartzites from Zverino.

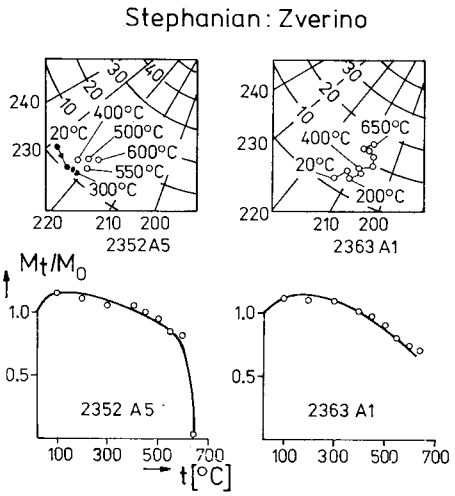


Fig. 21. Thermal demagnetization of Stephanian red quartzites from Zverino.

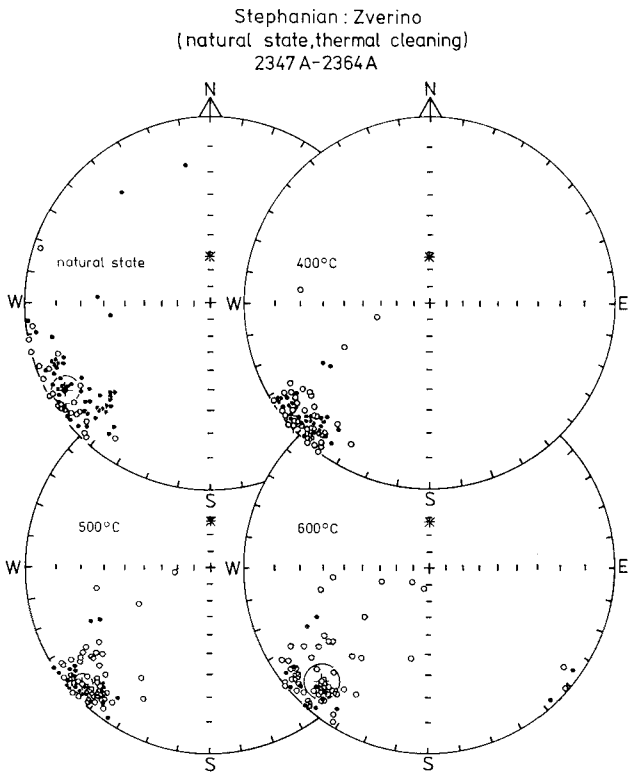


Fig. 22. Directions of NRMP and of RMP of thermally treated samples of Stephanian red quartzites from Zverino, cf. Tab. 1.

Table 7. Thermal cleaning of collections of Stephanian sediments from Bulgaria: Zverino. Sample No.: 2347A—2364A.

t °C	Mean directions of remanent magnetic polarization		α_{95}	k	N
	D	I			
Natural state	240·10°	5·83°	4·42°	15·05	73
400	228·19°	—1·24°	2·98°	32·83	71
500	227·46°	—4·48°	3·11°	30·31	71
600	222·73°	—10·32°	6·21°	8·34	71

a number of samples are magnetically isotropic. If the degree of anisotropy is considered for anisotropic samples according to the value of the ratio $P = \varkappa_1/\varkappa_3$ (where $\varkappa_1$ and $\varkappa_3$ are maximum and minimum susceptibility, respectively), then the sample 2197A displays the highest anisotropy in its natural state; $P = 1{\cdot}267$. The average value of P obtained for the whole set is $\bar{P} = 1{\cdot}0693 \pm 0{\cdot}0581$ ($N = 20$). The anisotropic samples (Nos. 2197A, 2198A, 2243A, 2248A, 2250A, 2258A, 2260A, 2268A and 2280A) are characterized by the value of $\bar{P} = 1{\cdot}0753 \pm 0{\cdot}0681$ ($N = 10$), what is a value slightly higher than that obtained for the whole group, but the difference between these two values is statistically insignificant. The nature of preferred orientation can suitably be determined using the coefficient $E = \varkappa_2^2/(\varkappa_1\varkappa_3)$, where $\varkappa_2$ is the intermediate susceptibility. Linear parallel and plane parallel orientation predominates if $E < 1$ and $E > 1$, respectively. The mean value of E for the entire set in natural state is $\bar{E} = 0{\cdot}986 \pm 0{\cdot}029$; for the anisotropic samples $\bar{E}' = 0{\cdot}989 \pm \pm 0{\cdot}029$. Consequently, linear parallel orientation predominates in both cases under natural state. The coefficient of $L = \varkappa_1/\varkappa_2$ is used to consider the degree of linear parallel orientation since the value obtained in the investigated group for anisotropic samples is $\bar{L} = 1{\cdot}044 \pm 0{\cdot}045$. The linear parallel orientation thus attains a value of $\sim 4{\cdot}4\%$.

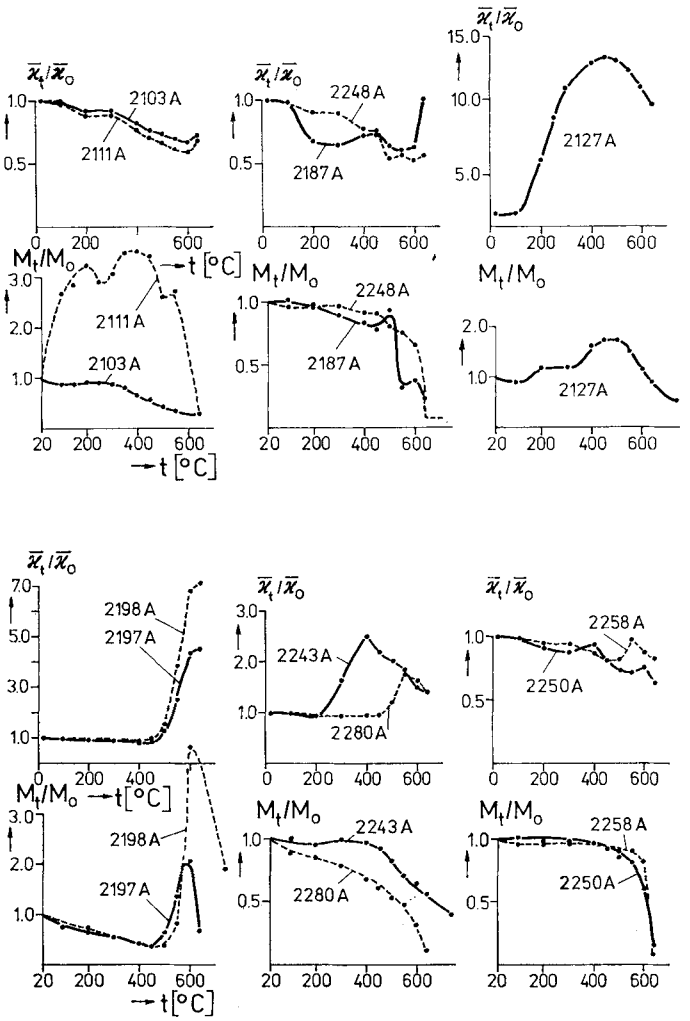
Considerable changes were noted in the values of both magnetic susceptibility and magnetic susceptibility anisotropy for the investigated samples as a result of thermal effects in the compensated Earth's magnetic field [during thermal cleaning of the natural remanent magnetic polarization, $M_t/M_0 = f(t)$]. The magnitude of changes in $\varkappa(t)/\varkappa_0$ [where $\varkappa(t)$ is the susceptibility measured at 20 °C after heating to the temperature t and $\varkappa_0$ is the susceptibility in natural state] is shown in Fig. 23a, b. The samples under study can be divided according to the nature of the changes

observed into two groups: one includes the samples for which $\varkappa(t)/\varkappa_0 > 1$ and the other, for $\varkappa(t)/\varkappa_0 < 1$. The three temperature domains could be distinguished at which $\varkappa(t)$ underwent greatest changes: ~200 °C, 400 °C and above 600 °C.

From Figs 23a,b it can also be inferred that the nature of changes in $\varkappa(t)/\varkappa_0$ responding to both phase and chemical changes is clearly reflected in the curves of magnetic moments, M_t/M_0.

Heating is also accompanied by changes in magnetic susceptibility anisotropy. Three groups can be differentiated:

i) isotropic samples within the whole interval of thermal treatment;



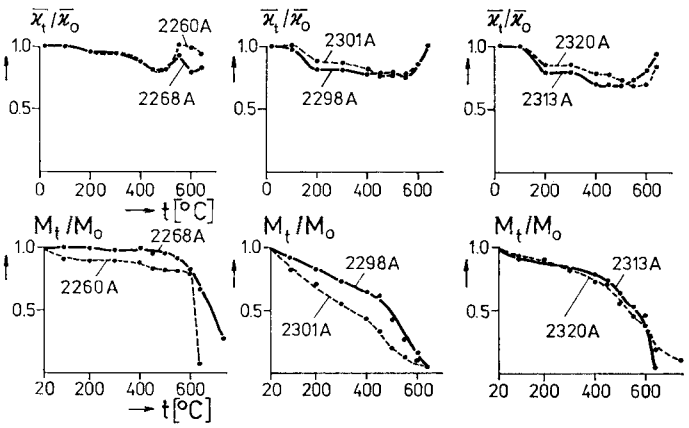
Obr. 23a

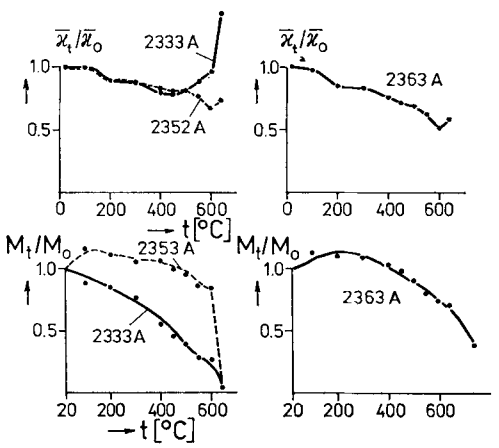
ii) anisotropic samples within the whole interval of thermal treatment;

iii) samples which are anisotropic within some thermal intervals but isotropic within others.

Accordingly, the first group comprises samples of Nos. 2103A, 2298A, 2301A, 2313A, 2320A, 2333A and 2352A; the second group comprises Nos. 2243A, 2248A, 2250A, 2258A, 2260A and 2258A; and the third group has Nos. 2127A, 2187A, 2197A, 2198A, 2280A ano 2363A.

Susceptibility was decreased by heating all the samples of the first group to a temperature of as much as 600 °C; a remarkable increase in susceptibility and a change to anisotropic susceptibility cannot sometimes be observed until the samples are





Obr. 23b

Fig. 23a, b. Temperature dependence of normalized mean volume magnetic susceptibility $\varkappa_t/\varkappa_0$ and of normalized remanent magnetic moment M_t/M_0 for different samples from Bulgaria, cf. Tab. 1.

heated to a temperature exceeding 600 °C and 640 °C. However, these changes do not have an essential influence upon the course of M_t/M_0 curves. This group includes isotropic samples; mean susceptibility attains the value of $\bar{\varkappa} = 68{\cdot}67 \times 10^{-6}$ [SI].

The second group contains samples which retain their magnetic susceptibility anisotropy. The value of mean susceptibility $\bar{\varkappa} = 209{\cdot}72 \times 10^{-6}$ [SI]. Once heated to a temperature of 200 °C, the specimens change their nature of preferred orientation; the linear parallel orientation (characterized by the value of $E < 1$) largely turns to plane parallel orientation ($E > 1$) and the lineation coefficients, L, are smaller than foliation coefficients, $F = \varkappa_2/\varkappa_3$. There is a very small dacrease in the value of $\overline{\varkappa(200\ °C)/\varkappa_0} = 0{\cdot}935 \pm 0{\cdot}024$, yet $\bar{L} = 1{\cdot}019 \pm 0{\cdot}05$ and $\bar{F} = 1{\cdot}027 \pm 0{\cdot}015$.

After heating at temperatures exceeding 400 °C, the specimens show a slight decrease in their susceptibility $\overline{\varkappa(400\ °C)/\varkappa_0} = 0{\cdot}870 \pm 0{\cdot}064$ (except sample of No. 2243A, for which $\varkappa(400\ °C)/\varkappa_0 = 2{\cdot}519$) and the prevailing plane parallel orientation retains its value: $\bar{F} = 1{\cdot}029 \pm 0{\cdot}016$; $\bar{L} = 1{\cdot}012 \pm 0{\cdot}008$ and $\bar{E} = 1{\cdot}016 \pm 0{\cdot}021$. The total degree of anisotropy $\overline{P(400\ °C)} = \varkappa_1/\varkappa_3 = 1{\cdot}041 \pm 0{\cdot}012$ became moderately lower [$\overline{P(20\ °C)} = 1{\cdot}055 \pm 0{\cdot}010$]. It can also be stated that the directions of main susceptibilities are approximately retained until such temperatures and the position of the susceptibility ellipsoid remains stable.

Essentially more remarkable changes in magnetic susceptibility and its anisotropy are produced by heating the specimens at 600 to 640 °C. The ratio of $\overline{\varkappa(640\ °C)/\varkappa_0} = 0{\cdot}883 \pm 0{\cdot}289$ stays rather invariable, but the scatter of values of $\varkappa(640\ °C)/\varkappa_0$ increased considerably. The character of preferred orientation, represented by the values of $\bar{L} = 1{\cdot}018 \pm 0{\cdot}014$; $\bar{F} = 1{\cdot}061 \pm 0{\cdot}017$, $\bar{P} = 1{\cdot}073 \pm 0{\cdot}018$ and $\bar{E} = 1{\cdot}049 \pm 0{\cdot}019$, supports the conclusions regarding the plane parallel arrangement of the ferrimagnetic matrix. However, the main axes of the susceptibility ellipsoid do not occupy the original position and change considerably.

The third group of the samples is characterized in its early stage by the following values: $\bar{L} = 1{\cdot}072 \pm 0{\cdot}069$; $\bar{F} = 1{\cdot}036 \pm 0{\cdot}025$; $\bar{P} = 1{\cdot}112 \pm 0{\cdot}094$; $\bar{E} = 0{\cdot}968 \pm 0{\cdot}047$ and $\varkappa_0 = 72{\cdot}87 \times 10^{-6}$ [SI]. If heated at 200 °C, the specimens display a slight decrease in the susceptibility value $\overline{\varkappa(200\ °C)/\varkappa_0} = 0{\cdot}869 \pm 0{\cdot}118$ [except for sample of No. 2127A for which $\varkappa(200\ °C)/\varkappa_0 = 5{\cdot}26$]. This is also accompanied by a slight change in the nature of anisotropy since $\bar{L} = 1{\cdot}068 \pm 0{\cdot}120$; $\bar{F} = 1{\cdot}027 \pm 0{\cdot}022$; $\bar{P} = 1{\cdot}094 \pm 0{\cdot}118$ and $\bar{E} = 0{\cdot}980 \pm 0{\cdot}088$. Except for sample of No. 2127A, these changes are statistically insignificant. At the same time, however, the originally isotropic samples of Nos. 2127A, 2187A and 2363A become anisotropic and the originally anisotropic samples of Nos. 2197A, 2198A and 2280A turn to isotorpic ones. Again, with the exception of sample of No. 2127A [showing $\overline{\varkappa(400\ °C)/\varkappa_0} = 12{\cdot}97$], most specimens are marked by a decrease in susceptibility when heated at 400 °C; hence $\overline{\varkappa(400\ °C)/\varkappa_0} = 0{\cdot}830 \pm 0{\cdot}088$; $\bar{L} = 1{\cdot}043 \pm 0{\cdot}047$; $\bar{F} = 1{\cdot}046 \pm 0{\cdot}050$; $\bar{P} = 1{\cdot}093 \pm 0{\cdot}101$ and $\bar{E} = 1{\cdot}003 \pm 0{\cdot}027$. This means that linear parallel orientation changes to plane parallel one and the degree of anisotropy

gradually lowers; $P = \varkappa_1/\varkappa_3$ as $P(20\,^\circ\mathrm{C}) = 1{\cdot}112$; $P(200\,^\circ\mathrm{C}) = 1{\cdot}094$; $P(400\,^\circ\mathrm{C}) = 1{\cdot}093$.

Considerable changes were due to heating the specimens at 640 °C : $\overline{\varkappa(640\,^\circ\mathrm{C})/\varkappa_0} = 4{\cdot}096 \pm 3{\cdot}705$; $\bar{L} = 1{\cdot}072 \pm 0{\cdot}059$; $\bar{F} = 1{\cdot}152 \pm 0{\cdot}122$; $\bar{P} = 1{\cdot}234 \pm 0{\cdot}145$ and $\bar{E} = 1{\cdot}078 \pm 0{\cdot}133$.

The only isotropic samples after heating at 640 °C are those of Nos. 2187A and 2363A. An increase in the susceptibility values can now be observed in all samples (except for No. 2363A) and preferred orientation assumes the nature of distinctly plane parallel arrangement. The degree of anisotropy, P, is high too. The main axes of the magnetic susceptibility ellipsoid strikingly change their position after heating all samples of the third group within the range of $0-360°$ and $0-90°$ for D and I, respectively. However, no regularities were observed during the process.

By comparing Figs 23a,b it can be seen that the samples increasing their magnetic susceptibility by thermal treatment have the main axes of the susceptibility ellipsoid occupying variable position and, at the same time, show unstable NRMP due to their chemical reworking during their geological history.

Slight changes (decrease) in the curves of $\varkappa_t/\varkappa_0$ after heating at 200 °C, at which

Table 8. Review of samples of magnetically isotropic (I) and anisotropic (A) sediments investigated during thermal treatment.

Sample No.	20 °C	200 °C	400 °C	500 °C	600 °C	640 °C
2103A	I	I	I	I	A	A
2111A	I	I	A	I	A	A
2127A	I	A	A	A	A	A
2187A	I	A	I	A	A	I
2197A	A	I	I	A	A	A
2198A	A	I	I	A	A	A
2243A	A	A	A	A	A	A
2248A	A	A	A	A	A	A
2250A	A	A	A	A	A	I
2258A	A	A	A	A	A	A
2260A	A	A	A	A	A	A
2268A	A	A	A	A	A	A
2280A	A	A	I	A	A	A
2298A	I	I	I	I	I	I
2301A	I	I	I	I	I	I
2313A	I	I	I	I	I	I
2320A	I	I	I	I	I	I
2333A	I	I	I	I	I	I
2352A	I	I	I	I	I	I
2363A	I	A	I	I	I	I

time $\varkappa$ decreases by 5–20% in all samples (except No. 2127A), can be explained by the release of water from Fe-hydroxides. The decrease in $\varkappa$ value caused by thermal treatment can be observed until the temperature of 400 to 450 °C is attained; at higher temperatures, however, magnetic susceptibility suddenly increases in practically all the samples by 20 to 400%. By comparing the $\varkappa_t/\varkappa_0$ curves with M_t/M_0 it is evident that the nature of the curves is similar in many characters, particularly in the samples of the third group. This can be explained by the fact that the $\varkappa_t/\varkappa_0$ curve qualitatively represents the course of phase and chemical changes also reflected in the M_t/M_0 curve trends.

The samples showing great changes in their susceptibility ($\varkappa_t/\varkappa_0 > 0{\cdot}2-0{\cdot}3$) possess also unstable NRMP whereas those with stable NRMP display only slight changes in $\varkappa_t$ until a temperature of about 500 °C is attained.

In general, it can be stated that thermal treatment makes all anisotropic specimens change their linear parallel orientation to plane parallel orientation. As far as the samples showing unstable NRMP and altered chemically during epigenesis are concerned, the anisotropy ellipsoid is highly unstable and changes can be observed in the directions of the main axes of the susceptibility tensor. The isotropic-anisotropic changes taking place during thermal treatment are shown for individual samples in Tab. 8.

5. PALAEOMAGNETIC PARAMETERS: DISCUSSION OF PRINCIPAL RESULTS

i) From the palaeomagnetic investigations of Triassic, Permian and Upper Carboniferous sediments in Bulgaria it was possible to reach a methodological conclusion that the samples of porous red sandstones were remagnetized because of the chemical alterations resulting from weathering. The remagnetized specimens showed values of the Curie temperature lower than those for haematite or mineral phase changes were observed during thermal demagnetization. The samples of palaeomagnetically unstable rocks showed changes of magnetic anisotropy during thermal treatments. Conversely, the red pelitic sediments containing a higher amount of clay fraction proved suitable for palaeomagnetic analyses after magnetic cleaning in a thermal field, with haematite always being a carrier of magnetization. The quartzites pigmented by haematite could also suitably be used for inferring palaeomagnetic directions. Such samples remained magnetically isotropic or anisotropic during laboratory thermal treatments.

ii) Basic palaeomagnetic data derived on Triassic, Lower Permian and Stephanian sediments of Bulgaria are given in Tab. 9. Palaeomagnetic directions and palaeomagnetic pole positions for the Triassic, Lower Permian and Upper Carboniferous are compatible with the values obtained for the tectonically stable North-European Platform. Consequently, palaeotectonic deformation effects as are known from the Alpine belt (cf. [12]) have not been observed in the region under study. The samples

Table 9. Triassic, Lower Permian, Upper Carboniferous palaeomagnetic directions and pole positions (Bulgaria) determined in this study. (Cleaning at 400 °C).

Sample Nos.	Age	Locality, lithology	Mean geogr. coordinates		Mean palaeomag. directions		α_{95}	k	N	Palaeomagnetic pole positions		Ovals of confidence	
			φ	λ	D	I				Palaeolatitude	Palaeolongitude	δ_m	δ_p
2272A—2346A cf. Tab. 4	Triassic	Tserovo, Bov, red beds	43·033°N	23·367°E	26·07°	39·35°	2·63°	23·33	129	60·04°N	148·85°E	3·15°	1·88°
2215A—2271A cf. Tab. 6	Lower Permian	Smolianovci, red beds	43·517°N	23·000°E	18·72°	27·15°	3·32°	19·13	100	56·75°N	168·46°E	3·61°	1·97°
2347A—2364A cf. Tab. 7	Upper Carboniferous (Stephanian)	Zverino, red beds	43·267°N	23·567°E	228·19°	—1·24°	2·98°	32·83	71	29·53°S	35·37°W	2·98°	1·49°

were taken from the marginal part of the Moesian Platform. The results thus far obtained support the conclusion that the Moesian Platform, if viewed in a megatectonic pattern, was not considerably rotated with respect to the North-European Platform during the Alpine orogeny.

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